

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVC503 - STRUCTURAL MECHANICS - II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

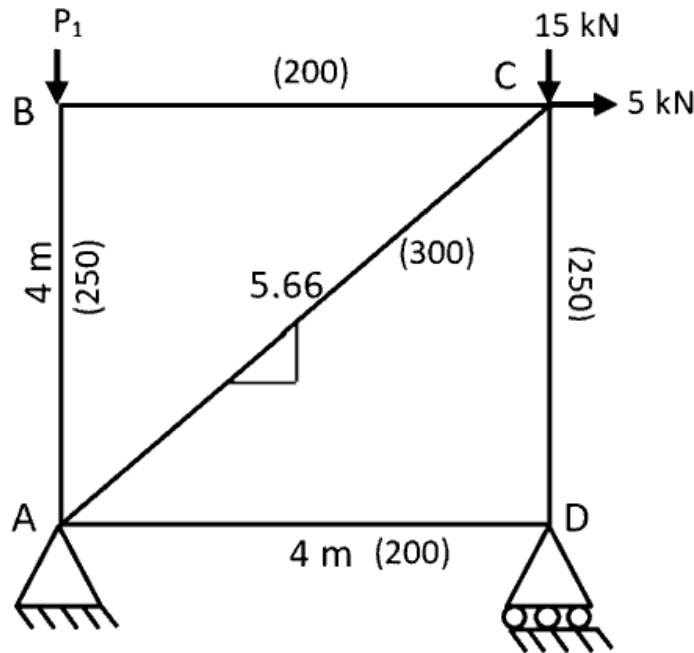
12

- 1 For determinate trusses external loads are acting at -----
a) Joints b) Members c) Both a and b d) None of these
- 2 Which one of the following is not the correct statement about Müller-Breslau principle?
(a) It is applicable when the material is within the elastic limit and obeys Hooke's law.
(b) It gives excellent experimental approach to determine the influence lines for indeterminate structures.
(c) It is applicable only for indeterminate structures.
(d) It gives analytical method to find ILD.
- 3 The influence line for a reaction in a simply supported beam is ----
a) parabola b) rectangle c) straight line d) hyperbola
- 4 In a suspension bridge, the cable is assumed to take the shape of a parabola under uniformly distributed load. The tension in the cable is maximum at:
(a) The center of the span (b) The supports (c) The quarter points of the span (d) None
- 5 A three-hinged arch is statically determinate because:
(a) It has three hinges (b) It has four reactions
(c) It has two hinges and one roller support (d) It has two hinges and one fixed support
- 6 In a suspension bridge, the shape of the cable under its own weight is a:
a) Parabola b) Catenary c) Circle d) Ellipse
- 7 The flexibility matrix method is used for the analysis of...
A) Statically determinate structures B) Statically indeterminate structures
C) Both statically determinate and indeterminate structures D) None
- 8 In a linear elastic structural element-----
a) Stiffness is directly proportional to stiffness
b) stiffness is equal to flexibility
c) Stiffness is inversely proportional to stiffness
d) Stiffness and flexibility are not related
- 9 Unit for stiffness is -----
a) N-m b) kN-mm c) N/m d) Both a & b
- 10 In finite element modeling every element connects to
a) 4 nodes b) 3 Nodes c) 2 Nodes d) Infinite No of Nodes

- 11 For truss analysis, which type of elements are used
 a) Triangle b) Bars c) Rectangle d) Parallelogram
- 12 Basic unknowns in the flexibility matrix method are -----
 a) Redundant forces b) Displacements of joints
 c) Both (a) & (b) d) None of these

Q2. Solve the following.

- A) Draw neat sketch of influence line diagrams (ILD) of Simply supported beam of span L for three cases: 6
 i) Reaction at Support A. ii) Shear Force at section B. iii) Bending Moment at section B.
- B) Determine the **vertical displacement** of the joint B in the pin jointed plane frame as shown in fig. below. The cross-section areas of the members in mm² are indicated alongside the members. Take $E = 2 \times 10^5 \text{ N/mm}^2$.



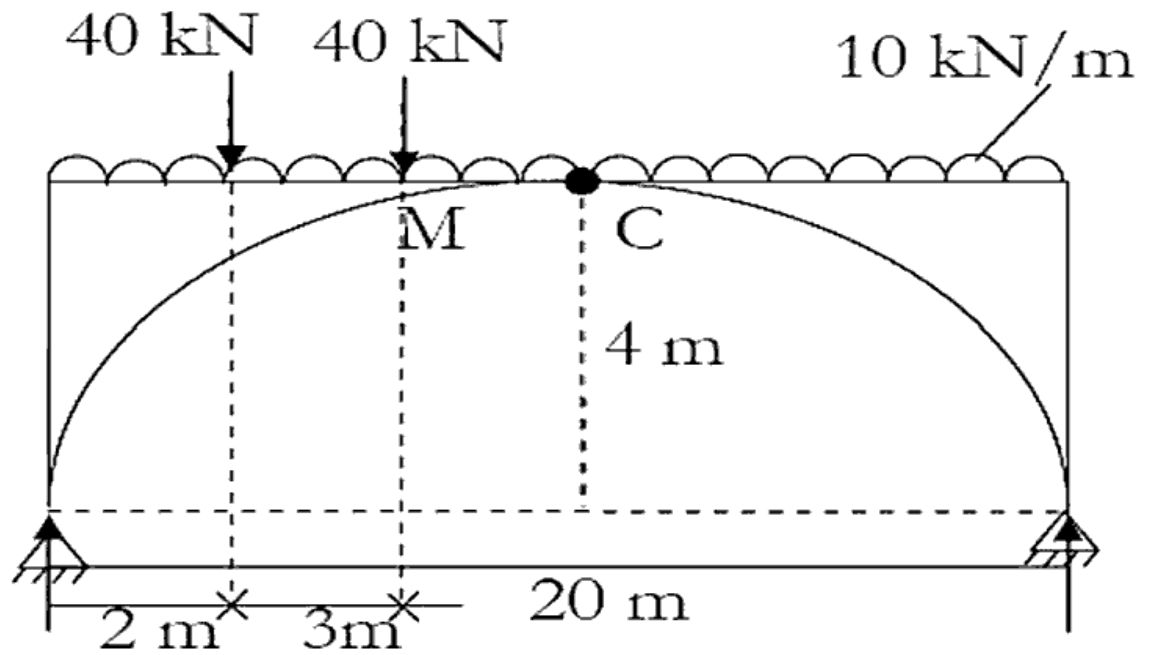
Q3. Solve the following.

- A) Explain in detail the components of suspension bridge with neat sketch. 6
- B) Explain difference between two hinged arch and three hinged arch with reference to its structural action. 6

Q4. Solve Any Two of the following.

- A) Write the stepwise procedure to solve indeterminate structure by using direct flexibility method of structural analysis. 6

- B) A symmetrical 3-hinged parabolic arch has a span of 20m. It carries UDL of intensity 10 kN/m over the entire span and two-point loads of 40 kN each at 2m and 5m from left support. Compute the reactions at the supports.



- C) Explain the concept of Muller Breslau's principle 6

Q5. Solve Any Two of the following.

- A) Define Stiffness & write the procedure/steps to be adopted to solve any problem by using 6 stiffness/displacement method of analysis.
- B) Develop the **stiffness matrix** for the cantilever beam of span L subjected to point load 'P' kN and clockwise 6 moment 'M' at free end.
- C) For the continuous beam shown in Q.3 C generate the flexibility matrix. 6

Q6. Solve Any Two of the following.

- A) Explain in detail the concept of 'shape function'. 6
- B) Explain in detail the concept of 'Pascal's Triangle'. 6
- C) Explain the different type of finite elements. 6

*** End ***